

Inductive Satisfiability Certificate for Universal Quantifiers & Uninterpreted Function Symbols



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A motivating example

$$f(0) = 0 \wedge \forall x \in \mathbb{Z}. f(x+1) = f(x) + 1$$

- ▶ satisfiable; model: $f(n) = n$ for all $n \in \mathbb{Z}$; **no finite model**
- ▶ current SMT techniques: MBQI fails; SyGuS needs a grammar with $\lambda x. x$
- ▶ the question: witness satisfiability *without* constructing f



a model of f : a value on *every* $n \in \mathbb{Z}$ — an infinite object

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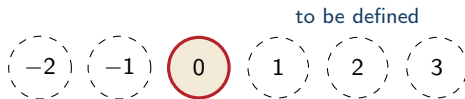


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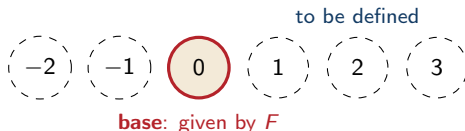


a model of f : a value on every $n \in \mathbb{Z}$ — an infinite object

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$$\underbrace{f(0) = 0}_F \wedge \forall x \in \mathbb{Z}. \underbrace{f(x+1) = f(x) + 1}_Q$$

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A motivating example: the certificate

ground model I : $f(0) = 0$
well-order $\preceq$: $0, 1, -1, 2, -2, \dots$

$x \geq 0$: write $f(x+1)$, $f(x+1) := f(x) + 1$



$x < 0$: write $f(x)$, $f(x) := f(x+1) - 1$

why two regions?

the rule $f(x+1) := f(x) + 1$, at $x = -1$, writes $f(0)$
 $f(0)$: fixed by F , never written

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instantiation	reads	writes
$x = 0$	$f(0) = 0$	$f(1) := f(0) + 1 = 1$
$x = 1$	$f(1) = 1$	$f(2) := f(1) + 1 = 2$
$x = -1$	$f(0) = 0$	$f(-1) := f(0) - 1 = -1$

each step reads only cells fixed earlier
induction along $\preceq$

Problem statement and Goal

Formulas $F \wedge \forall \bar{x}. Q$ with F, Q quantifier-free; uninterpreted function symbols, arguments linear in $\bar{x}$; no nesting — UFLIA

Problem satisfiability: does a model of $F \wedge \forall \bar{x}. Q$ exist?

Difficulty **undecidable** (Horbach et al., 2017) — no algorithm for all instances

Goal **SAT** or **UNSAT** — even without a finite model, without an explicit function

Unsat answers are well supported; sat answers are not

- ▶ SMT solvers: strong at **Unsat** — evidence: proofs
- ▶ for Sat, the common methods:
- ▶ finite model search (e.g. *MBQI*)
- ▶ explicit function synthesis (e.g. *SyGuS*)

Our target

Sat problems with **no finite model, no explicit function**

Why sat answers matter

conjecture φ : check $\neg\varphi$

unsat: φ proved

sat: φ disproved — the model: a counterexample

unknown: neither a proof nor a counterexample

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family	what it needs
finite model search	a finite model
MBQI: Ge, de Moura; Reynolds et al.	
explicit function synthesis	a grammar containing the answer
SyGuS: Kuncak et al., Hozzová et al.	
array procedures	the index arithmetic restricted
Bradley et al., Habermehl et al., Alberti et al.	
model templates	a fixed shape for f — here: the rule is
Brown et al.	synthesized

this talk: the instances where the others fail — the satisfiability certificate

Preliminaries: cells

f : not a function — a *store*:

- ▶ a *cell*: a ground application $f(\bar{a})$, $\bar{a} \in \mathbb{Z}^k$
- ▶ a partial interpretation: an assignment of values to cells
- ▶ each instantiation of Q : *reads* the cells it mentions, *writes* further cells

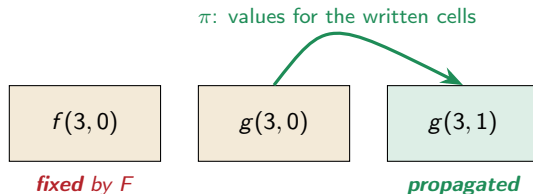


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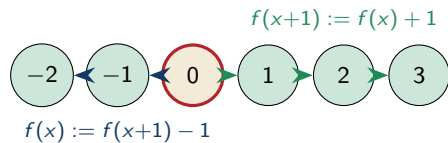


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Definition: the satisfiability certificate (informal version)

the motivating example, generalized — a *satisfiability certificate* for $F \wedge \forall \bar{x}. Q$ over $\mathbb{Z}$:

- (i) a *ground model*: a model of F — the starting point
- (ii) a *well-order* $\preceq$: an order where induction works
- (iii) *propagated cells*: the cells it may write — never a cell of F , never a cell read earlier
- (iv) a **satisfiability propagator**: whatever the values so far — some values for the written cells make it true



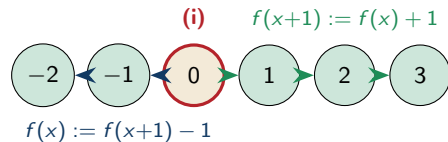
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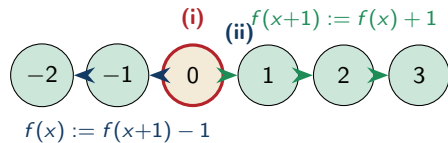
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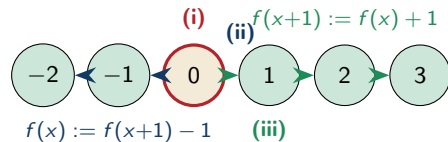
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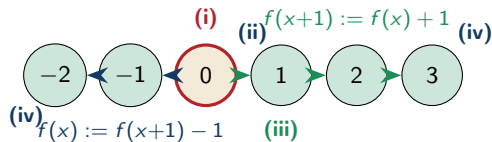
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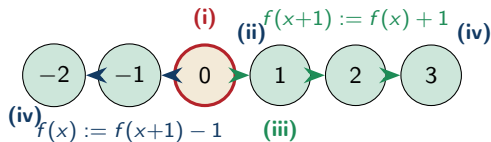
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Theorem (soundness)

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The certificate, formally

$F \wedge \forall \bar{x}. Q$ over $\mathbb{Z}$; $\Gamma_I(\varphi)$: the relevant cells of φ under I ; $update(J, \pi)$: J with the cells in $\text{dom } \pi$ set to π :

- (i) a ground model $I \models F$
- (ii) a well-order $\preceq$ on $\mathbb{Z}^{|\bar{x}|}$
- (iii) for each $\bar{z}$: propagated cells $X_{\bar{z}} \subseteq \Gamma_I(Q[\hat{\bar{z}}])$ with $X_{\bar{z}} \cap \Gamma_I(F) = \emptyset$ and $X_{\bar{z}} \cap \bigcup_{\bar{z}' \prec \bar{z}} \Gamma_I(Q[\hat{\bar{z}}']) = \emptyset$
- (iv) a **satisfiability propagator**: for every J with $J|_{\Gamma_I(F)} = I|_{\Gamma_I(F)}$ and every $\bar{z}$ there is $\pi : X_{\bar{z}} \rightarrow \mathbb{Z}$ with $update(J, \pi) \models Q[\hat{\bar{z}}]$

Theorem (soundness)

Every formula that has a satisfiability certificate is satisfiable.

A more complex example

$$f(3, a) \geq 4 + g(0, b) \wedge b > 0 \quad \wedge \quad \forall x, y. [f(x, y) < x + g(x, y) \wedge g(x, y) = g(x, y + b) + 2b]$$

the certificate: I ; the order $\preceq$;
 $X_{\bar{z}}$ = column 2; propagator = column 3

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ground model $I \models F$:

$$a = 0, \quad b = 1, \quad f(3, 0) = 4, \quad g(0, 1) = 0$$

its cells $\Gamma_I(F)$ — the cells of F : **not writable**

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order $\preceq$	propagated cells $X_{\bar{z}}$	propagator
$x = 3, y = 0$	$g(3, 0), g(3, 1)$	$g(3, 0) := 2, g(3, 1) := 0$
$y \geq 0, (x, y) \neq (3, 0)$	$f(x, y), g(x, y + 1)$	$g(x, y + 1) := g(x, y) - 2, f(x, y) := x + g(x, y) - 1$
$y < 0$	$f(x, y), g(x, y)$	$g(x, y) := g(x, y + 1) + 2, f(x, y) := x + g(x, y + 1) + 1$

every instantiation covered;

no written cell in $\Gamma_I(F)$ or in cells read earlier

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Checkable conditions in linear integer arithmetic

the informal certificate, items (iii) and (iv) — as first-order formulas: regions $\text{cond}_1, \dots, \text{cond}_m$ of $\mathbb{Z}^k$ (e.g. $x \geq 0$, $x < 0$); per region: a pivot set $\mathcal{S} \subseteq \mathcal{T}$ — the terms whose cells propagation writes; $\bar{u}$: read, $\bar{v}$: written.

clash_F (iii): pivot cells avoid the cells of F	quantifier-free
coll_Q (iii): pivot cells at $\bar{z}$ avoid cells read at $\bar{z}' \prec \bar{z}$	quantifier-free
prop_Q (iv): every read valuation admits pivot values satisfying Q	$\forall \bar{u} \exists \bar{v}$ over $\mathbb{Z}$

The congruences — an example first

$\forall x. f(2x) = f(x) + 1$: at $x = 0$, $f(2x)$ and $f(x)$ — the same cell, separate variables
equal arguments $\Rightarrow$ equal values: $u_{f(\bar{t})} = u_{f(\bar{t}')} \text{ (read)}, \quad v_{f(\bar{t})} = v_{f(\bar{t}')} \text{ (written)}$

in general, for the record: $\text{cong}_{\text{non-pivot}}^S \equiv \bigwedge_{f(\bar{t}), f(\bar{t}') \in \mathcal{T} \setminus \mathcal{S}} (\bar{t} = \bar{t}' \Rightarrow u_{f(\bar{t})} = u_{f(\bar{t}')})$,
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they hold — for an unsatisfiable formula.

Coinciding terms: the missing conjunct

$$\top \wedge \forall x. f(2x) = f(x) + 1$$

- ▶ at $x = 0$: the pivot term $f(2x)$ and the non-pivot term $f(x)$ **coincide**
- ▶ prop_Q never relates $v_{f(2x)}$ and $u_{f(x)}$: $\forall u \exists v. v = u + 1$ valid
- ▶ one cell, two variables in the abstraction: $v \neq u$ permitted

The missing conjunct: a congruence for coinciding terms

$$\text{cong}_{\text{coincide}}^S \equiv \bigwedge_{f(\bar{t}) \in S} \bigwedge_{f(\bar{t}') \in T \setminus S} (\bar{t} = \bar{t}' \Rightarrow v_{f(\bar{t})} = u_{f(\bar{t}')})$$

add $\text{cong}_{\text{coincide}}^S$ to the $\exists \bar{v}$ -conjunct of prop_Q

at $x = 0$: $v = u$ from $\text{cong}_{\text{coincide}}^S$, but $v = u + 1$ from $Q(S)$ — the condition **fails**

soundness refined: F together with $\text{clash}_F \wedge \text{coll}_Q \wedge \text{prop}_Q$
with $\text{cong}_{\text{coincide}}^S$ for every region, satisfiable $\Rightarrow$ a certificate exists

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add $\text{cong}_{\text{coincide}}^S$ to the $\exists \bar{v}$ -conjunct of prop_Q

at $x = 0$: $v = u$ from $\text{cong}_{\text{coincide}}^S$, but $v = u + 1$ from $Q(S)$ — the condition **fails**

soundness refined: F together with $\text{clash}_F \wedge \text{coll}_Q \wedge \text{prop}_Q$
with $\text{cong}_{\text{coincide}}^S$ for every region, satisfiable $\Rightarrow$ a certificate exists

The procedure: the idea

$$\Psi = F \wedge \bigwedge_{\text{regions } i} \bigvee_{\text{pivot sets } \mathcal{S}_i} \Phi_i^{\mathcal{S}_i}$$

- ▶ $\Phi_i^{\mathcal{S}_i}$: the checkable conditions — clash_F , coll_Q , prop_Q — for region i and pivot set $\mathcal{S}_i$
- ▶ $\Psi \text{ sat} \Rightarrow$ a certificate exists (the model of Ψ , the pivot sets) $\Rightarrow$ the formula is satisfiable

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The procedure: the loop

the query can fail, even for satisfiable formulas — $f(-10)$, $f(10)$ of F stand in the way:

$$f(-10) = 0 \wedge f(10) = c \wedge \forall x. f(x) = f(x-1) + 1 \quad \text{— clash}_F: \text{ no pivot set survives}$$

Input: $F \wedge \forall \bar{x}. Q$, regions $\text{cond}_1, \dots, \text{cond}_m$,
order $\preceq$

$R_0 := \emptyset$; $i := 0$; $\Phi_{\text{base}} := F$

while Φ_{base} is not UNSAT do

$\Psi := \Phi_{\text{base}}$ and, for every region $R_i \setminus R_0$:
 some pivot set makes the
 conditions true

 if Ψ is SAT then return SAT

$R_0 := R_0 \cup \text{shell}_i(\mathbb{Z}^k)$

$\Phi_{\text{base}} := F$ and every instance in R_0

$i := i+1$

od

return UNSAT

► repair: replace F and Q by
 $F' = F \wedge \bigwedge_{x \in [-10, 10]} f(x) = f(x-1) + 1$

$$Q' = \forall x. x \notin [-10, 10] \Rightarrow f(x) = f(x-1) + 1$$

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► **UNSAT** $\Leftrightarrow F$ plus finitely many instances of Q : Unsatisfiable

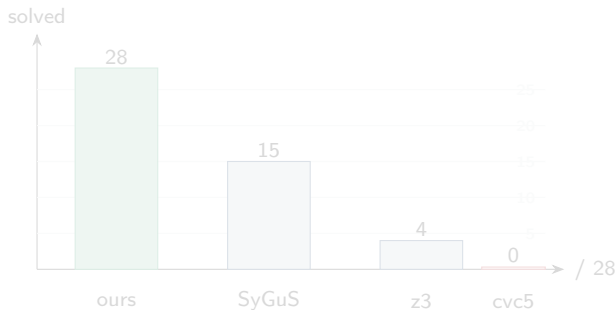
Soundness, completeness, evaluation

Guarantees

- ▶ soundness: both answers sound
- ▶ completeness (relative): a model with pivot sets — coll_Q , prop_Q hold in it; finitely many pivot cells in $\Gamma_I(F) \Rightarrow$ **returns SAT**
- ▶ termination: finitely many instances refute the formula

Prelim result

- ▶ 28 handcrafted satisfiable UFLIA problems; functions of arity 1
- ▶ ours: Python on z3; baselines z3, cvc5, cvc5-SyGuS



problems solved on the unbounded variants;

ours: 28/28, runtime insensitive to the bound 10^c

cvc5: 54s at $c = 3$ on $f(4) = 7 \wedge \forall x. f(x+1) = f(x) + 1$, timeout for $c \geq 4$

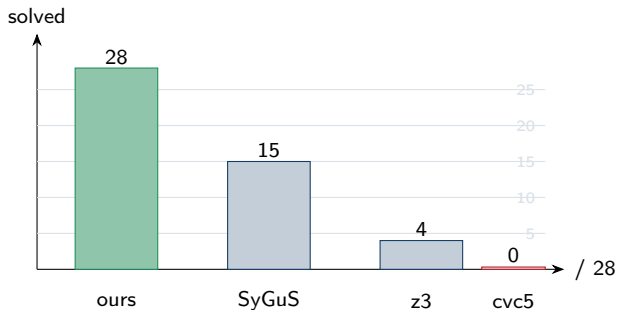
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Limitations and further work

- ▶ **no certificate exists** — the main limitation; two simple examples:
 - not enough cells for propagation: $f(0) = 0 \wedge \forall x, y. f(x) = f(y)$ — two quantified variables, f of arity one
 - circular dependencies: $\forall x, y. f(x, y) = f(y, x)$; with a guard ($x \geq y \Rightarrow f(x, y) = f(y, x)$): three regions
- ▶ the fragment: no nested uninterpreted functions; regions and order supplied by the user
- ▶ 28 handcrafted problems so far; larger corpora pending

Further work

weaker propagation conditions; stronger forms of induction;
ML for the region search

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The message of this talk

A finite, independently checkable object can witness satisfiability — even when no finite model exists.

in three words: **certify, don't construct**



Stefan Nugraha

a.k.a. Anggha Nugraha

github.com/MarekDanco/f-definitions

the arXiv preprint is outdated -- these slides follow the current version

with Stefan Ratschan, Mikoláš Janota, Marek Dančo — thank you!

Backup: a costly base region

$$f(10000) > 0 \wedge \forall x. f(x+1) = f(x) + 1$$

- ▶ at $x = 10^4 - 1$: the pivot cell $f(x+1)$ — fixed by F : clash_F
- ▶ grounding fixes $f(10^4 - 1)$ — the clash cascades downwards
- ▶ the base interval: one shell per iteration, $\sim 10^4$ expansions — sound, but slow; future work